

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282194

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Graus; Jonathon P. et al.

VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING

Abstract

A damping control system for a vehicle having a suspension located between a plurality of ground engaging members and a vehicle frame are disclosed. The vehicle including at least one adjustable shock absorber having an adjustable damping characteristic.

Inventors: Graus; Jonathon P. (Wyoming, MN), LeMay; Adam J. (Wyoming, MN), Anderson; Justin T. (Wyoming, MN), Olson; Kyle W. (Wyoming, MN)

Applicant: Polaris Industries Inc. (Medina, MN)

Family ID: 96894490

Appl. No.: 19/064339

Filed: February 26, 2025

Related U.S. Application Data

us-provisional-application US 63561627 20240305

us-provisional-application US 63677480 20240731

Publication Classification

Int. Cl.: B60G17/016 (20060101); B62M27/02 (20060101)

U.S. Cl.:

CPC B60G17/0164 (20130101); B62M27/02 (20130101); B60G2300/07 (20130101); B60G2400/0512 (20130101); B60G2400/106 (20130101); B60G2500/104 (20130101); B62M2027/026 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/561,627, filed Mar. 5, 2024, entitled “VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING,” and the benefit of U.S. Provisional Patent Application Ser. No. 63/677,480, filed Jul. 31, 2024, entitled “VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING,” the contents of which are expressly incorporated herein by reference.

FIELD OF DISCLOSURE

[0002] The present disclosure relates to improved suspension characteristics for a vehicle and in particular to systems and methods of damping control, such as compression and/or rebound damping, for shock absorbers.

BACKGROUND OF THE DISCLOSURE

[0003] Adjustable shock absorbers are known. Systems and methods for controlling one or more adjustable characteristics of adjustable shock absorbers are disclosed in US Published Patent Application No. 2016/0059660 (filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL) and US Published Application 2018/0141543 (filed Nov. 17, 2017, titled VEHICLE HAVING ADJUSTABLE SUSPENSION), assigned to the present assignee, the entire disclosures of which are expressly incorporated by reference herein.

SUMMARY OF THE DISCLOSURE

[0004] In an exemplary embodiment of the present disclosure, a recreational vehicle is provided.

[0005] Some embodiments of the present disclosure provide a recreational vehicle. The recreational vehicle includes: a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, and at least one sensor positioned on the recreational vehicle and configured to provide one or more of acceleration information, pitch information, or position information to a controller. Each of the plurality of suspensions couple a ground engaging member, from the plurality of ground engaging members, to the frame. The plurality of suspensions include a plurality of adjustable shock absorbers. The controller is operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers. The controller is configured to: receive, from the at least one sensor, the one or more of acceleration information, pitch information, or position information; determine, based on the one or more of acceleration information, pitch information, or position information, to adjust damping of at least one of the plurality of adjustable shock absorbers; and provide, to the at least one of the plurality of adjustable shock absorbers, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

[0006] Some embodiments of the present disclosure provide a method of adjusting one or more adjustable shock absorbers of a vehicle. The method includes: receiving, from at least one sensor of the vehicle, acceleration information and pitch information; determining, based on the acceleration information and pitch information, to adjust damping of at least one of the one or more adjustable shock absorbers; and providing, to the at least one of the one or more adjustable shock absorbers, one or more commands to cause an adjustment of a damping characteristic of the at least one of the one or more adjustable shock absorbers.

[0007] Some embodiments of the present disclosure provide a recreational vehicle. The recreational vehicle includes: a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, and at least one sensor positioned on the recreational vehicle, the at least one sensor configured to provide acceleration information and pitch information to a controller. Each of the plurality of suspensions couple a ground engaging member, from the plurality of ground engaging members, to the frame and

include an adjustable shock absorber of a plurality of adjustable shock absorbers. The controller is operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers. The controller is configured to: receive, from the at least one sensor, the acceleration information and the pitch information; calculate an event score based on the acceleration information and the pitch information; compare the calculated event score to a threshold; and control at least one adjustable shock absorber of the plurality of adjustable shock absorbers to adjust a damping characteristic of the at least one adjustable shock absorber based on the comparison.

[0008] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects, features, and/or advantages of examples will be set forth in part in the following description and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The foregoing aspects and many additional features of the present system and method will become more readily appreciated and become better understood by reference to the following detailed description when taken in conjunction with the accompanying drawings, where:

[0010] FIG. 1 shows a representative view of components of a vehicle of the present disclosure having a suspension with a plurality of continuous damping control shock absorbers and a plurality of sensors integrated with a controller of the vehicle;

[0011] FIG. 2 shows an adjustable damping shock absorber coupled to a vehicle suspension;

[0012] FIG. 3 shows an x-axis, a y-axis, and a z-axis for a vehicle, such as an ATV;

[0013] FIG. 4 shows a representative view of an exemplary power system for the vehicle of FIG. 1;

[0014] FIG. 5 shows a representative view of an exemplary controller of the vehicle of FIG. 1;

[0015] FIG. 6 shows a first, perspective view of an exemplary vehicle;

[0016] FIG. 7 shows a second, perspective view of the exemplary vehicle of FIG. 6;

[0017] FIG. 8 shows a side view of the exemplary vehicle of FIG. 6;

[0018] FIG. 9 shows a bottom view of the exemplary vehicle of FIG. 6;

[0019] FIG. 10 shows an exemplary control system for controlling the damping of one or more shock absorbers;

[0020] FIG. 11 shows an example rear loading event, according to some embodiments of the present disclosure;

[0021] FIG. 12 shows an example flowchart describing the operation of a suspension controller during a rear loading event, according to some embodiments of the present disclosure;

[0022] FIG. 13 shows an example rear unloading event, according to some embodiments of the present disclosure;

[0023] FIG. 14 shows an example flowchart describing the operation of a suspension controller during a rear unloading event, according to some embodiments of the present disclosure;

[0024] FIG. 15 shows an example of the suspension controller adjusting the adjustable shock absorbers for the vehicle during rear loading/unloading events;

[0025] FIG. 16 shows an exemplary schematic block diagram illustrating an example of logic components in the suspension controller;

[0026] FIG. 17 illustrates a representative view of a stabilizer bar of the vehicle of FIG. 1; and

[0027] FIG. 18 illustrates a view of another exemplary vehicle.

DETAILED DESCRIPTION OF EMBODIMENTS

[0028] For the purposes of promoting an understanding of the principles of the present disclosure,

reference will now be made to the embodiments illustrated in the drawings, which are described below. The embodiments disclosed below are not intended to be exhaustive or limited to the precise form disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings.

[0029] Referring now to FIG. 1, the present disclosure relates to a vehicle **10** having a suspension system **11** located between a plurality of ground engaging members **12** and a vehicle frame **14**. Exemplary ground engaging members **12** include wheels, skis, guide tracks, treads or other suitable devices for supporting the vehicle relative to the ground. The suspension typically includes springs **16** and shock absorbers **18** coupled between the ground engaging members **12** and the frame **14**. The springs **16** may include, for example, coil springs, leaf springs, air springs or other gas springs. The air or gas springs **16** may be adjustable. See, for example, U.S. Pat. No. 7,950,486, entitled VEHICLE (filed Jun. 6, 2008), and assigned to the current assignee, the entire disclosure of which is incorporated herein by reference.

[0030] The adjustable shock absorbers **18** are often coupled between the vehicle frame **14** and the ground engaging members **12** through an A-arm linkage **70** (See FIG. 2) or other type linkage. Springs **16** are also coupled between the ground engaging members **12** and the vehicle frame **14**. FIG. 2 illustrates an adjustable shock absorber **18** mounted on an A-arm linkage **70** having a first end pivotably coupled to the vehicle frame **14** and a second end pivotably coupled to A-arm linkage **70** which moves with wheel **12**. A damping control activator **74** is coupled to controller **20** by one or more wires **76**. An exemplary damping control activator is an electronically controlled valve which is activated to increase or decrease the damping characteristics of adjustable shock absorber **18**.

[0031] In one embodiment, the adjustable shock absorbers **18** include solenoid valves mounted at the base of the shock body or internal to a damper piston of the shock absorber **18**. The stiffness of the shock is increased or decreased by introducing additional fluid to the interior of the shock absorber, removing fluid from the interior of the shock absorber, and/or increasing or decreasing the ease with which fluid can pass from a first side of a damping piston of the shock absorber to a second side of the damping piston of the shock absorber. In another embodiment, the adjustable shock absorbers **18** include a magnetorheological fluid internal to the shock absorber **18**. The stiffness of the shock is increased or decreased by altering a magnetic field experienced by the magnetorheological fluid. Additional details on exemplary adjustable shocks are provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, entitled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, and assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0032] In one embodiment, a spring **16** and shock **18** are located adjacent each of the ground engaging members **12**. In an all-terrain vehicle (ATV), for example, a spring **16** and an adjustable shock **18** are provided adjacent each of the four wheels **12**. In a snowmobile, for example, one or more springs **16** and one or more adjustable shocks **18** are provided for each of the two front skis and the rear tread. For example, the snowmobile can have a front adjustable shock **18** for the front skis and a rear adjustable shock **18** for the rear skis. Some manufacturers offer adjustable springs **16** in the form of either air springs or hydraulic preload rings. These adjustable springs **16** allow the operator to adjust the ride height on the go. However, a majority of ride comfort comes from the damping provided by shock absorbers **18**.

[0033] In an illustrated embodiment, controller **20** provides signals and/or commands to adjust damping characteristics of the adjustable shocks **18**. For example, the controller **20** provides signals to adjust damping of the shocks **18** in a continuous or dynamic manner. In other words, adjustable shocks **18** may be adjusted to provide differing compression damping, rebound damping, or both. In one embodiment, adjustable shocks **18** include a first controllable valve to adjust compression damping and a second controllable valve to adjust rebound damping. In another embodiment, adjustable shocks include a combination valve which controls both compression damping and

rebound damping.

[0034] In an illustrated embodiment of the present disclosure, an operator interface **22** is provided in a location easily accessible to the driver operating the vehicle. For example, the operator interface **22** is either a separate user interface mounted adjacent the driver's seat on the dashboard or integrated onto a display within the vehicle. Operator interface **22** includes user input devices to allow the driver or a passenger to manually adjust shock absorber **18** damping during operation of the vehicle based on road conditions that are encountered or to select a preprogrammed active damping profile for shock absorbers **18** by selecting a ride mode. Examples of techniques for controlling a damping characteristics of an adjustable shock absorber of a vehicle are provided in U.S. Pat. No. 10,406,884, entitled ADJUSTABLE VEHICLE SUSPENSION SYSTEM (filed Jun. 9, 2017), assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0035] Exemplary input devices for operator interface **22** include levers, buttons, switches, soft keys, and other suitable input devices. Operator interface **22** may also include output devices to communicate information to the operator. Exemplary output devices include lights, displays, audio devices, tactile devices, and other suitable output devices. In another illustrated embodiment, the user input devices are on a steering wheel, handle bar, or other steering control of the vehicle **10** to facilitate actuation of the damping adjustment. For example, a physical switch may be located on the steering wheel, handle bar, or other steering control of the vehicle **10**. In some examples, a display may be provided on or next to the operator interface **22** or integrated into a dashboard display of vehicle **10** to display information related to the compression and/or rebound damping characteristics.

[0036] As explained in further detail below, controller **20** receives kinematic measurements from one or more vehicle condition sensors **40** and adjusts the damping characteristics of the adjustable shocks **18** accordingly. The operator may independently adjust front and rear shock absorbers **18** to adjust the ride characteristics of the vehicle **10**. In certain embodiments, each of the shocks **18** is independently adjustable so that the damping characteristics of the shocks **18** are changed from one side of the vehicle **10** to another. The damping response of the shock absorbers **18** can be changed in a matter of milliseconds to provide nearly instantaneous changes in damping for potholes, dips in the road, hills, or other driving conditions. Additionally, and/or alternatively, controller **20** may independently adjust the damping characteristics of front and/or rear shocks **18**. An advantage, among others, of adjusting the damping characteristics of the front and/or rear shocks **18** is that the vehicle **10** may be able to operate more efficiently in rough terrain and/or improve a rider's experience.

[0037] The controller **20** communicates with (e.g., provides, transmits, receives, and/or obtains) multiple vehicle condition sensors **40**. For example, a wheel accelerometer **25** is coupled adjacent each ground engaging member **12**. The controller **20** communicates with each of the accelerometers **25**. For instance, the accelerometers **25** may provide information indicating movement of the ground engaging members and the suspension components **16** and **18** as the vehicle traverses different terrain. Further, the controller **20** may communicate with additional vehicle condition sensors **40**, such as a vehicle speed sensor **26**, a steering sensor **28**, a chassis supported accelerometer **30**, a chassis supported gyroscope **31**, an inertial measurement unit (IMU) **37** (shown on FIG. **10**), a physical switch **820** (shown on FIG. **10**), and other sensors which monitor one or more characteristics of vehicle **10**.

[0038] Accelerometer **30** is illustratively a three-axis accelerometer supported on the chassis of the vehicle **10** to provide information indicating acceleration forces of the vehicle **10** during operation. In some instances, accelerometer **30** is located at or close to a center position (e.g., a center of gravity position) of vehicle **10**. In other instances, the accelerometer **30** is located at a position that is not near the center of gravity of the vehicle **10**. In the exemplary vehicle **200** illustrated in FIGS. **6-9**, the chassis accelerometer **30** is located along a longitudinal centerline plane **122** of vehicle

200. The x-axis, y-axis, and z-axis for a vehicle **10**, illustratively an ATV, are shown in FIG. 3.

[0039] Gyroscope **31** is illustratively a three-axis gyroscope supported on the chassis to provide indications of inertial measurements, such as roll rates, pitch rates, and/or yaw rates, of the vehicle during operation. In one embodiment, accelerometer **30** is not located at a center of gravity of vehicle **10** and the readings of gyroscope **31** are used by controller **20** to determine the acceleration values of vehicle **10** at the center of gravity of vehicle **10**. In one embodiment, accelerometer **30** and gyroscope **31** are integrated into the controller **20**, such as a suspension controller **86**.

[0040] In some examples and referring to FIG. 10, an IMU, such as the IMU **37** is supported on the chassis to provide indications of the inertial measurements, including the angular rate and/or the acceleration forces, of the vehicle **10** during operation. The IMU **37** may include the functionalities of the accelerometer **30** and/or the gyroscope **31**. As such, in some instances, the accelerometer **30** and/or the gyroscope **31** are optional and might not be included in the vehicle **10**. In other instances, the vehicle **10** may include the gyroscope **31** and the accelerometer **30** instead of the IMU **37**.

[0041] The controller **20** may also communicate with additional vehicle condition sensors **40**, such as a brake sensor **32**, a throttle position sensor **34**, a wheel speed sensor **36**, and/or a gear selection sensor **38**.

[0042] Referring to FIG. 4, one embodiment of a driveline torque management system **50** of vehicle **10** is illustrated. Driveline torque management system **50** controls the amount of torque exerted by each of ground engaging members **12**. Driveline torque management system **50** provides a positive torque to one or more of ground engaging members **12** to power the movement of vehicle **10** through a power system **60**. Driveline torque management system **50** further provides a negative torque to one or more of ground engaging members **12** to slow or stop a movement of vehicle **10** through a braking system **75**. In one example, each of ground engaging members **12** has an associated brake of braking system **75**.

[0043] Power system **60** includes a prime mover **62**. Exemplary prime movers **62** include internal combustion engines, two stroke internal combustion engines, four stroke internal combustion engines, diesel engines, electric motors, hybrid engines, and other suitable sources of motive force. To start the prime mover **62**, a power supply system **64** is provided. The type of power supply system **64** depends on the type of prime mover **62** used. In one embodiment, prime mover **62** is an internal combustion engine and power supply system **64** is one of a pull start system and an electric start system. In one embodiment, prime mover **62** is an electric motor and power supply system **64** is a switch system which electrically couples one or more batteries to the electric motor.

[0044] A transmission **66** is coupled to prime mover **62**. Transmission **66** converts a rotational speed of an output shaft **61** of prime mover **62** to one of a faster rotational speed or a slower rotational speed of an output shaft **63** of transmission **66**. It is contemplated that transmission **66** may additionally rotate output shaft **63** at the same speed as output shaft **61**.

[0045] In the illustrated embodiment, transmission **66** includes a shiftable transmission **68** and a continuously variable transmission (“CVT”) **71**. In one example, an input member of CVT **71** is coupled to prime mover **62**. An input member of shiftable transmission **68** is in turn coupled to an output member of CVT **71**. In one embodiment, shiftable transmission **68** includes a forward high setting, a forward low setting, a neutral setting, a park setting, and a reverse setting. The power communicated from prime mover **62** to CVT **71** is provided to a drive member of CVT **71**. The drive member in turn provides power to a driven member through a belt. Exemplary CVTs are disclosed in U.S. Pat. Nos. 3,861,229; 6,176,796; 6,120,399; 6,860,826; and 6,938,508, the disclosures of which are expressly incorporated by reference herein. The driven member provides power to an input shaft of shiftable transmission **68**. Although transmission **66** is illustrated as including both shiftable transmission **68** and CVT **71**, transmission **66** may include only one of shiftable transmission **68** and CVT **71**. Further, transmission **66** may include one or more additional components.

[0046] Transmission **66** is further coupled to at least one differential **73** which is in turn coupled to at least one ground engaging members **12**. Differential **73** may communicate the power from transmission **66** to one of ground engaging members **12** or multiple ground engaging members **12**. In an ATV embodiment, one or both of a front differential and a rear differential are provided. The front differential powering at least one of two front wheels of the ATV and the rear differential powering at least one of two rear wheels of the ATV. In a side-by-side vehicle embodiment having seating for at least an operator and a passenger in a side-by-side configuration, one or both of a front differential and a rear differential are provided. The front differential powering at least one of two front wheels of the side-by-side vehicle and the rear differential powering at least one of multiple rear wheels of the side-by-side vehicle. In one example, the side-by-side vehicle has three axles and a differential is provided for each axle. An exemplary side-by-side vehicle **200** is illustrated in FIGS. **6-9**.

[0047] In one embodiment, braking system **75** includes anti-lock brakes. In one embodiment, braking system **75** includes active descent control and/or engine braking. In one embodiment, braking system **75** includes a brake and in some embodiments a separate parking brake. Braking system **75** may be coupled to any of prime mover **62**, transmission **66**, differential **73**, and ground engaging members **12** or the connecting drive members therebetween. Brake sensor **32**, in one example, monitors when braking system **75** is applied. In one example, brake sensor **32** monitors when a user actuatable brake input, such as brake pedal **232** (see FIG. **7**) in vehicle **200**, is applied.

[0048] Referring to FIG. **5**, controller **20** has at least one associated memory **76**. Controller **20** provides the electronic control of the various components of vehicle **10**. Further, controller **20** is operatively coupled to a plurality of vehicle condition sensors **40** as described above, which monitor various parameters of the vehicle **10** or the environment surrounding the vehicle **10**. Controller **20** performs certain operations (e.g., provides commands) to control one or more subsystems of other vehicle components. In certain embodiments, the controller **20** forms a portion of a processing subsystem including one or more computing devices having memory, processing, and communication hardware. Controller **20** may be a single device or a distributed device, and the functions of the controller **20** may be performed by hardware and/or as computer instructions on a non-transitory computer readable storage medium, such as memory **76**.

[0049] As illustrated in the embodiment of FIG. **5**, controller **20** is represented as including several controllers. These controllers may each be single devices or distributed devices or one or more of these controllers may together be part of a single device or distributed device. The functions of these controllers may be performed by hardware and/or as computer instructions on a non-transitory computer readable storage medium, such as memory **76**.

[0050] In one embodiment, controller **20** includes at least two separate controllers which communicate over a network **78**. In one embodiment, network **78** is a CAN network. Details regarding an exemplary CAN network are disclosed in U.S. patent application Ser. No. 11/218,163, filed Sep. 1, 2005, the disclosure of which is expressly incorporated by reference herein. Of course any suitable type of network or data bus may be used in place of the CAN network. In one embodiment, two wire serial communication is used for some connections.

[0051] Referring to FIG. **5**, controller **20** includes an operator interface controller **80** which controls communication with an operator through operator interface **22**. A prime mover controller **82** controls the operation of prime mover **62**. A transmission controller **84** controls the operation of transmission system **66**.

[0052] A suspension controller **86** controls adjustable portions of suspension system **11**. Exemplary adjustable components include adjustable shocks **18**, adjustable springs **16**, and/or configurable stabilizer bars. Additional details regarding adjustable shocks, adjustable springs, and configurable stabilizer bars is provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, entitled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, and assigned to the present assignee, the entire disclosure of which is expressly

incorporated by reference herein.

[0053] Communication controller **88** controls communications between a communication system **90** of vehicle **10** and remote devices, such as other vehicles, personal computing devices, such as cellphones or tablets, a centralized computer system maintaining one or more databases, and other types of devices remote from vehicle **10** or carried by riders of vehicle **10**. In one embodiment, communication controller **88** of vehicle **10** communicates with paired devices over a wireless network. An exemplary wireless network is a radio frequency network utilizing a BLUETOOTH protocol. In this example, communication system **90** includes a radio frequency antenna. Communication controller **88** controls the pairing of devices to vehicle **10** and the communications between vehicle **10** and the remote device. In one embodiment, communication controller **88** of vehicle **10** communicates with remote devices over a cellular network. In this example, communication system **90** includes a cellular antenna and communication controller **88** receives and sends cellular messages from and to the cellular network. In one embodiment, communication controller **88** of vehicle **10** communicates with remote devices over a satellite network. In this example, communication system **90** includes a satellite antenna and communication controller **88** receives and sends messages from and to the satellite network. In one embodiment, vehicle **10** is able to communicate with other vehicles **10** over a Radio Frequency mesh network and communication controller **88** and communication system **90** are configured to enable communication over the mesh network. An exemplary vehicle communication system is disclosed in U.S. patent application Ser. No. 15/262,113, filed Sep. 12, 2016, entitled VEHICLE TO VEHICLE COMMUNICATIONS DEVICE AND METHODS FOR RECREATIONAL VEHICLES, and assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

[0054] A steering controller **102** controls portions of a steering system **104**. In one embodiment, steering system **104** is a power steering system and includes one or more steering sensors **28** (shown in FIG. **1**). Exemplary sensors and electronic power steering units are provided in U.S. patent application Ser. No. 12/135,107, assigned to the assignee of the present application, titled VEHICLE, docket PLR-06-22542.02P, the disclosure of which is expressly incorporated by reference herein. A vehicle controller **92** controls lights, loads, accessories, chassis level functions, and other vehicle functions. A ride height controller **96** controls the preload and operational height of the vehicle. In one embodiment, ride height controller controls springs **16** to adjust a ride height of vehicle **10**, either directly or through suspension controller **86**. In one example, ride height controller **96** provides more ground clearance in a comfort ride mode compared to a sport ride mode.

[0055] An agility controller **100** controls a braking system of vehicle **10** and the stability of vehicle **10**. Control methods of agility controller **100** may include integration into braking circuits (ABS) such that a stability control system can improve dynamic response (vehicle handling and stability) by modifying the shock damping in conjunction with electronic braking control.

[0056] In one embodiment, controller **20** either includes a location determiner **110** and/or communicates via network **78** to a location determiner **110**. The location determiner **110** determines a current geographical location of vehicle **10**. An exemplary location determiner **110** is a GPS unit which determines the position of vehicle **10** based on interaction with a global satellite system.

[0057] Referring to FIGS. **6-9**, an exemplary side-by-side vehicle **200** is illustrated. Vehicle **200**, as illustrated, includes a plurality of ground engaging members **12**. Illustratively, ground engaging members **12** are wheels **204** and associated tires **206**. As mentioned herein, one or more of ground engaging members **12** are operatively coupled to power system **60** (see FIG. **4**) to power the movement of vehicle **200** and braking system **75** to slow movement of vehicle **200**.

[0058] Referring to the illustrated embodiment in FIG. **6**, a first set of wheels, one on each side of vehicle **200**, generally correspond to a front axle **208**. A second set of wheels, one on each side of vehicle **200**, generally correspond to a rear axle **210**. Although each of front axle **208** and rear axle

210 are shown having a single ground engaging member **12** on each side, multiple ground engaging members **12** may be included on each side of the respective front axle **208** and rear axle **210**. As configured in FIG. 6, vehicle **200** is a four wheel, two axle vehicle.

[0059] Referring to FIG. 9, wheels **204** of front axle **208** are coupled to a frame **212** of vehicle **200** through front independent suspensions **214**. Front independent suspensions **214** in the illustrated embodiment are double A-arm suspensions. Other types of suspensions systems may be used for front independent suspensions **214**. The wheels **204** of rear axle **210** are coupled to frame **212** of vehicle **200** through rear independent suspensions **216**. Other types of suspensions systems may be used for rear independent suspensions **216**.

[0060] Returning to FIG. 6, vehicle **200** includes a cargo carrying portion **250**. Cargo carrying portion **250** is positioned rearward of an operator area **222**. Operator area **222** includes seating **224** and a plurality of operator controls. In the illustrated embodiment, seating **224** includes a pair of bucket seats. In one embodiment, seating **224** is a bench seat. In one embodiment, seating **224** includes multiple rows of seats, either bucket seats or bench seats or a combination thereof. Exemplary operator controls include a steering wheel **226**, a gear selector **228**, an accelerator pedal **230** (see FIG. 7), and a brake pedal **232** (see FIG. 7). Steering wheel **226** is operatively coupled to the wheels of front axle **208** to control the orientation of the wheels relative to frame **212**. Gear selector **228** is operatively coupled to the shiftable transmission **68** to select a gear of the shiftable transmission **68**. Exemplary gears include one or more forward gears, one or more reverse gears, and a park setting. Accelerator pedal **230** is operatively coupled to prime mover **62** to control the speed of vehicle **200**. Brake pedal **232** is operatively coupled to brake units associated with one or more of wheels **204** to slow the speed of vehicle **200**.

[0061] Operator area **222** is protected with a cage **240**. Referring to FIG. 6, side protection members **242** are provided on both the operator side of vehicle **200** and the passenger side of vehicle **200**. In the illustrated embodiment, side protection members **262** are each a unitary tubular member.

[0062] In the illustrated embodiment, cargo carrying portion **250** includes a cargo bed **234** having a floor **256** and a plurality of upstanding walls. Floor **256** may be flat, contoured, and/or comprised of several sections. Portions of cargo carrying portion **250** also include mounts **258** which receive an expansion retainer (not shown). The expansion retainers which may couple various accessories to cargo carrying portion **250**. Additional details of such mounts and expansion retainers are provided in U.S. Pat. No. 7,055,454, to Whiting et al., filed Jul. 13, 2004, titled "Vehicle Expansion Retainers," the entire disclosure of which is expressly incorporated by reference herein.

[0063] Front suspensions **214A** and **214B** each include a shock absorber **260**, respectively. Similarly, rear suspensions **216A** and **216B** each include a shock absorber **262**. In one embodiment each of shock absorbers **260** and shock absorbers **262** are electronically adjustable shocks **18** which are controlled by a controller **20** of vehicle **200**.

[0064] Additional details regarding vehicle **200** are provided in U.S. Pat. Nos. 8,827,019 and 9,211,924, assigned to the present assignee, the entire disclosures of which are expressly incorporated by reference herein. Other exemplary recreational vehicles include ATVs, utility vehicles, snowmobiles, other recreational vehicles designed for off-road use, on-road motorcycles, and other suitable vehicles (e.g., FIG. 31).

[0065] FIG. 10 shows an exemplary control system **300** for controlling the damping of shock absorbers **18**. In some instances, the control system **300** may be included in the vehicle **10** and/or the vehicle **200** described above. For example, the suspension controller **86** may communicate with (e.g., receive and/or provide) one or more entities (e.g., sensors, devices, controllers, and/or subsystems) from the vehicle **10** and/or **200** described above. Additionally, and/or alternatively, the vehicle **10** and **200** may be the same vehicle (e.g., the vehicle **200** may include entities from vehicle **10**, such as the suspension controller **86**).

[0066] Additionally, the controller **20** may include a suspension controller **86** as described above in

FIG. 5. The suspension controller **86** may communicate with the plurality of vehicle condition sensors **40** as described above. Further, the suspension controller **86** may provide information (e.g., one or more commands) to each of the adjustable shock absorbers **18a**, **18b**, **18c**, and **18d**. For example, the suspension controller **86** may provide commands to adjust the compression damping characteristic and/or the rebound damping characteristic for the adjustable shock absorbers **18a**, **18b**, **18c**, and **18d**.

[0067] While exemplary sensors, devices, controllers, and/or subsystems are provided in FIG. **10**, additional exemplary sensors, devices, controllers, and/or subsystems used by the suspension controller **86** to adjust shock absorbers **18** are provided in US Published Patent Application No. 2016/0059660 (filed Nov. 6, 2015), entitled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, and US Published Application 2018/0141543 (filed Nov. 17, 2017), titled VEHICLE HAVING ADJUSTABLE SUSPENSION, both assigned to the present assignee and the entire disclosures of each expressly incorporated by reference herein.

[0068] The illustrative control system **300** is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the present disclosure. Neither should the illustrative control system **300** be interpreted as having any dependency or requirement related to any single entity or combination of entities illustrated therein. Additionally, various entities depicted in FIG. **10**, in embodiments, may be integrated with various ones of the other entities depicted therein (and/or entities not illustrated). For example, the suspension controller **86** may be included within the controller **20**, and may communicate with one or more vehicle condition sensors **40** as described above. The functionalities of the suspension controller **86** and/or other entities in control system **300** will be described below.

[0069] In some examples, the control system **300** can be used to control the damping of shock absorbers **18**, such as to reduce the effects of a rear loading event and/or a rear unloading event. In some examples, the control system **300** may be used to control the damping of shock absorbers **18**, such as to reduce the effect and an inclined and/or declined terrain traversed by the vehicle. In some examples, the rear loading event may be called a “G-Out” event in which the vehicle experiences rearward pitch motion while under high vertical G-loading. In some examples, the rear unloading event may be called a “Donkey Kick” event in which the vehicle experiences forward pitch motion while the vehicle is unloading from high vertical G-loading. In some examples, the control system **300** may be used to reduce the effects of a rear loading event to prevent a vehicle (e.g., vehicle **10** and/or vehicle **200**) from bottoming out of rear shocks. In some examples, the vehicle might experience a rear loading event where rear shocks of the vehicle bottom out because the vehicle is driven into an incline (e.g., a hill). For example, when the front of the vehicle traverses the incline it creates high vertical G-loading and then when the rear traverses the incline the rear shocks can bottom out due to the high vertical G-loading and rearward pitch motion. In some examples, vertical G-loading is a measure of the force experienced by the vehicle due to acceleration in the vertical direction, where 1G is equivalent to the force of gravity on Earth's surface.

[0070] In some examples, a rear loading event and a rear unloading event are related because the initial bottoming out of a vehicle onto the face of an incline and using the travel of the vehicle's suspension system (e.g., a spring/jounce bumper) can contribute to causing the rear end of the vehicle to kick upward into the air and/or forward, such as in a rear unloading event. For example, the bottoming out of a vehicle can refer to one or more adjustable shock absorbers (e.g., shock absorbers **18**) of a suspension system (e.g., the suspension system **11**) reaching the limits (or relatively near to the limits) of its travel or compression. In some examples, the bottoming out can be caused due to a sudden increase in load or a significant bump or dip in a surface along which the vehicle is travelling. The bottoming out can cause the vehicle to sustain a sudden jolt or impact, potentially putting strain on the suspension components. Further, in some examples, the sudden jolt (e.g., an increase in vertical acceleration) or impact can cause significant energy to be stored in the

springs of the suspension system (e.g., coil over spring, jounce bumper, tire), which can then result in a rear unloading event, such as when the energy is subsequently released. However, the rebound damping described herein can slow the release of this stored energy therefore reducing and/or eliminating the rear unloading event. During a rear loading event, systems described herein can reduce the energy stored in the springs of the suspension system by controlling compressing damping during the rear loading event.

Rear Loading Event (G-Out Event)

[0071] In some examples, the control system **300** enables a fast-acting functionality that momentarily stiffens up shock absorbers on a vehicle (e.g., rear shock absorbers), such as in situations where the vehicle is in a rear loading event, loading a vehicle's suspension into the face of a jump under compression, or generally if there is a relatively large amount of pitch motion. This functionality may thus result in a vehicle with an improved rider experience because the vehicle can have improved performance on relatively rough terrain.

[0072] FIG. **11** illustrates an example of a rear loading event, such as for vehicle **200** or vehicle **10**. In some examples, to reduce the effects of a rear loading event (e.g., where a rear of the vehicle may bottom out), the control system **300** may detect a change in vertical acceleration and change in pitch rate indicating that the vehicle is transitioning onto the face of an incline (e.g., a hill or other graded terrain) and/or landing a jump on the front axle of a vehicle with the rear axle then rapidly moving down. In examples, the control system **300** is configured to detect higher than normal G-loading and a rearward pitch rate (e.g., an increased vertical force or acceleration on the rear axle of a vehicle). In some examples, the change in vertical acceleration may be an increase or a decrease (e.g., depending on the point of reference). In some examples, the change in pitch rate may be an increase or a decrease (e.g., depending on the point of reference). In some examples, the control system **300** may detect a change in position of one or more components of the vehicle **200**. For example, the control system **300** may detect a change in position of one or more adjustable shock absorbers of the vehicle **200**. Other examples of vehicle components of which the control system **300** may monitor positioning to determine whether a rear loading event is occurring should be recognized by those of ordinary skill in the art.

[0073] In some examples, after detecting the changes in one or more of vertical acceleration, pitch rate, or vehicle component positioning, the control system **300** may then transmit a signal to enable an adjustment of a damping characteristic of one or more adjustable shock absorbers of the vehicle. In some examples, transmitting the signal may result in or cause the adjustment of the damping characteristic. In some examples, the adjustment of a damping characteristic (e.g., change in compression damping of one or more rear adjustable shock absorbers) may be implemented over a period of time (e.g., such that the change in rear compression is not implemented all at once). In some examples, the change in rear compression may be implemented all at once. In some examples, by changing (e.g., adding) rear compression to the vehicle, the control system **300** may prevent the rear from bottoming out (or, as another example, reduce the extent to which the rear uses compression travel) due to the terrain into which the vehicle is being driven and pitch motion of the vehicle. In some examples, adding the rear compression can enable improved control of vehicle pitch motion, thus helping to control the vehicle more easily. In some examples, the improved vehicle control may be particularly beneficial in whoops terrain (e.g., off-roading terrain with sudden drops, unexpected obstacles, and/or uneven surfaces) and/or in situations where an operator of the vehicle gets the vehicle pitching forward and back.

[0074] In some examples, the change in vertical acceleration, vehicle component positioning, and/or change in pitch rate are compared to one or more thresholds to determine whether to implement an adjustment of a damping characteristic to one or more adjustable shock absorbers of the vehicle. Therefore, by comparing the changes to such thresholds, mechanisms provided herein may ensure that adjustments to damping characteristics of adjustable shock absorbers are not made during relatively normal driving conditions, but rather only during rear loading events. In some

examples, the vertical acceleration is multiplied by the pitch rate to determine a rear loading event score and/or magnitude. In some examples, the rear loading event score is compared to a threshold to determine whether to implement an adjustment of a damping characteristic to one or more adjustable shock absorbers of the vehicle. Additional and/or alternative calculations, which use and/or are based on the changes in acceleration, vehicle component positioning, and/or pitch rate, to calculate the rear loading event score, should be recognized by those of ordinary skill in the art. [0075] In some examples, the rear loading event score, change in acceleration, change in vehicle component positioning, and/or change in pitch rate may be compared to a look-up table to determine whether to adjust a damping characteristic of one or more adjustable shock absorbers of the vehicle. In some examples, the comparison to the look-up table may be used to determine what type of damping characteristic (e.g., compression damping and/or rebound damping) to adjust. In some examples, the comparison to the look-up table may be used to determine how much to adjust the damping characteristic of the one or more adjustable shock absorbers of the vehicle.

[0076] In some examples, the thresholds to which the changes in vertical acceleration, vehicle component positioning, and/or pitch rate are compared may be calibrated. For example, the thresholds may be calibrated based on and/or correspond to a type of vehicle (e.g., vehicle model), mode of vehicle (e.g., ECO mode, sport mode, work mode, two-wheel drive, four-wheel drive, etc.), operator preference, manufacture preference, historical data of vehicle component positioning, acceleration, and/or pitch information (e.g., vertical acceleration and pitch rate) collected over a duration of time (e.g., hours, days, months years), or other factors which should be recognized by those of ordinary skill in the art. Similarly, in some examples, the look-up table may be calibrated. For example, the look-up table may be calibrated based on and/or correspond to a type of vehicle (e.g., vehicle model), mode of vehicle (e.g., ECO mode, sport mode, work mode, two-wheel drive, four-wheel drive, etc.), operator preference, manufacture preference, historical data of vehicle component positioning, acceleration, and/or pitch information (e.g., vertical acceleration and pitch rate) collected over a duration of time (e.g., hours, days, months years), or other factors which should be recognized by those of ordinary skill in the art.

[0077] In some examples, the control system **300** functions continuously, such that adjustments to damping characteristics of one or more shock absorbers of the vehicle are made smoothly, rather than as a binary on/off function. In some examples, the vertical acceleration and/or change in pitch rate are received from an IMU, such as the IMU **37** of FIG. **10**. In some examples, the vehicle component positioning measurements are received from a shock absorber position sensor (e.g., such as when measuring the position of shock absorbers). In some examples, the control system **300** may be in communication with other types of sensors, such as wheel acceleration sensors, shock acceleration sensors, axle sensors, camera sensors, radar sensors, or any other sensors recognized by those of ordinary skill in the art which may be used to detect when vehicle suspension is going to compress or unload. Accordingly, in some examples, the control system **300** can transmit one or more signals to enable adjustments to one or more adjustable shock absorbers, in addition to or independent of kinematic sensors of a vehicle. Examples of cameras/radars which may be used in accordance with aspects of the present disclosure are provided in U.S. patent application Ser. No. 18/592,643, entitled VEHICLE (filed Mar. 1, 2024), which is incorporated by reference herein in its entirety. Further, in some examples, terrain detection can be implemented, such as using terrain detection techniques described in PCT/US2024/012271, entitled TERRAIN DETECTION AND METHODS OF USE THEREOF (filed Jan. 19, 2024), which is incorporated by reference herein in its entirety. For example, one of ordinary skill in the art will recognize in view of PCT/US2024/012271 how a LIDAR sensor or other type of vision sensor can be used to provide input to the control system **300** for adjusting characteristics of the shock absorbers of the vehicle.

[0078] FIG. **12** shows an example flowchart describing the operation of the suspension controller **86** during a rear loading event. The flowchart illustrates an example method or process **400**. In

some examples, the method **400** is a method of adjusting one or more adjustable shock absorbers. As will be described in more detail below, by adjusting the compression damping of one or more adjustable shock absorbers **18** due to a rear loading event, the suspension controller **86** may increase the efficiency of vehicle **10** by preventing the vehicle **10** from bottoming out.

[0079] In operation, at step **402**, the suspension controller **86** may receive information (e.g., inputs) from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive (e.g., retrieve and/or obtain) information (e.g., data packets and/or signals indicating sensor readings) from the one or more sensors, devices, and/or subsystems. In some instances, the suspension controller **86** may receive information indicating the x, y, and/or z-axis acceleration from the chassis accelerometer **30** and/or IMU **37**. For example, referring back to FIG. **3**, the chassis accelerometer **30** and/or IMU **37** may measure the x, y, and/or z-axis acceleration values for the vehicle **10**, and may provide the acceleration values to the suspension controller **86**. Further, the suspension controller **86** may receive information indicating a pitch rate of the vehicle, such as from the IMU **37**. Still further, the suspension controller **86** may receive information indicating a position of a component of the vehicle, such as from a shock absorber positioning sensor and/or a positioning sensor configured to monitor the position of another vehicle component.

[0080] At step **406**, the suspension controller **86** may determine whether a rear loading event is occurring. For example, the suspension controller **86** may determine whether a rear loading event is occurring based on the z-axis acceleration value (i.e., vertical acceleration) and pitch rate of the vehicle. For instance, the suspension controller **86** may compare the z-axis acceleration and pitch rate values with one or more thresholds (e.g., one or more pre-defined, pre-programmed, and/or user-defined thresholds) to determine whether the vehicle **10** is in a rear loading event. If the z-axis acceleration value and pitch rate value are greater than the one or more thresholds, the method **400** may move to step **408**. If the z-axis acceleration value and pitch rate value are less than the thresholds, the method **400** may move back to step **402** and repeat.

[0081] In some examples, the suspension controller **86** may use any of a plurality of different techniques for comparing the z-axis acceleration and pitch rate to thresholds. In some examples, the z-axis acceleration is multiplied by the pitch rate to determine a rear loading event score and/or magnitude. In some examples, the rear loading event score is compared to a threshold to determine whether to implement an adjustment of a damping characteristic to one or more adjustable shock absorbers of the vehicle. Additional and/or alternative calculations, which use and/or are based on the changes in acceleration and pitch rate, to calculate the rear loading event score, should be recognized by those of ordinary skill in the art.

[0082] In some examples, the rear loading event score, change in acceleration, and/or change in pitch rate may be compared to a look-up table to determine whether to adjust a damping characteristic of one or more adjustable shock absorbers of the vehicle. In some examples, the comparison to the look-up table may be used to determine what type of damping characteristic (e.g., compression damping and/or rebound damping) to adjust. In some examples, the comparison to the look-up table may be used to determine how much to adjust the damping characteristic of the one or more adjustable shock absorbers of the vehicle.

[0083] At step **408**, the suspension controller **86** may determine a rear loading event modifier for one or more of the adjustable shock absorbers **18**. For example, the suspension controller **86** may provide information (e.g., one or more commands) to the adjustable shock absorbers **18** to increase and/or gradually increase the compression damping. By increasing the compression damping of the adjustable shock absorbers **18**, the vehicle **10** may increase the energy absorption capability of the shock absorbers to thus prevent or reduce the extent to which the rear of the vehicle bottoms out and/or an amount of suspension travel used.

[0084] At step **410**, the suspension controller **86** may determine whether a rear loading event has occurred and/or is done/completed. For example, similar to step **406**, the suspension controller **86**

may determine whether the rear loading event is done based on the z-axis acceleration value and/or pitch rate of the vehicle. For instance, the suspension controller **86** may compare the z-axis acceleration value and/or pitch rate with one or more thresholds (e.g., one or more pre-defined, pre-programmed, and/or user-defined thresholds) to determine whether the vehicle **10** is no longer in a rear loading event. In some examples, the suspension controller **86** may use the same thresholds as in step **406**.

[0085] At step **412**, the suspension controller **86** may execute a normal condition modifier. For example, the suspension controller **86** may adjust the compression damping back to normal (e.g., back to the pre-rear loading event position). For example, the suspension controller may adjust and/or gradually adjust the compression damping back to the normal (e.g., the pre-rear loading event setting).

[0086] In some examples, the method **400** terminates at step **412**. Alternatively, in some examples, the method **400** returns to step **402**, from step **412**, such as to provide a continuous loop of vehicle information (e.g., acceleration information and/or pitch information) to determine whether, and by how much, to adjust a damping characteristic of one or more adjustable shock absorbers. Thus, while method **400** depicts operation **408** as executing a rear loading event modifier and operation **412** as executing a normal condition modifier, it will be appreciated that, in some instances, similar aspects may be used to gradually adjust compression damping of a vehicle between such modifiers (e.g., gradually and/or as a combination thereof).

Rear Unloading Event (Donkey Kick Event)

[0087] FIG. **13** illustrates an example of a rear unloading event, such as for vehicle **200** or vehicle **10**. In some examples, to reduce the effects of a rear unloading event (e.g., where the rear end of a vehicle gets launched up and forward into the air, relative to the front end of the vehicle), the control system **300** may detect a change in vertical acceleration and change in pitch rate indicating that the vehicle is transitioning from an incline (e.g., of a hill or other graded terrain) into a decline. The rear unloading event is an unloading and/or forward pitch motion due to a vertical loading that has been removed from the vehicle **200**. In some examples, the change in pitch rate may be an increase or a decrease (e.g., depending on the point of reference).

[0088] In some examples, after detecting the changes in vertical acceleration and pitch rate, the control system **300** may then transmit a signal to enable an adjustment of a damping characteristic of one or more adjustable shock absorbers of the vehicle. In some examples, transmitting the signal may result in or cause the adjustment of the damping characteristic. In some examples, the adjustment of a damping characteristic (e.g., change in rebound damping of one or more rear adjustable shock absorbers) may be implemented over a period of time (e.g., such that the change in rear rebound damping is not implemented all at once). In some examples, the change in rear rebound damping may be implemented all at once. In some examples, by changing (e.g., adding) rear rebound damping to the vehicle, the control system **300** may slow down or prevent the rear end of the vehicle from being launched upward and/or forward relative to the front end of the vehicle. In some examples, a rear unloading event is characterized by a transition from high vertical G-loading to low vertical G-loading, while having a forward pitch motion and pitch acceleration, such as when front tires leave the terrain or the terrain begins to fall away from the vehicle (see FIG. **13**, as an example).

[0089] In some examples, the vertical acceleration is run through a low-pass filter (e.g., at 0.25 hertz). In some examples, running the vertical acceleration through a low-pass filter gives a delayed vertical acceleration signal. In some examples, the low-pass filtering is helpful, such as in cases where the vertical acceleration is dropping due to the vehicle becoming airborne. In some examples, the low-pass filter also prevents mechanisms from inadvertently triggering. In some examples, prior to performing calculations using the vertical acceleration and/or pitch acceleration, mechanisms provided herein verify that the vertical acceleration was/is greater than 1G and that the pitch acceleration is in a positive rotational direction (e.g., as illustrated by the rotational arrows

pointing clockwise in FIG. 13). In some examples, thresholds may be calibrated or otherwise configured, such that, for example, the acceleration threshold is different than 1G. In some examples, the low-pass filter is providing a phase delayed vertical acceleration signal that is representative of being previously “vertically loaded,” such that when the phase delayed vertical acceleration signal is coupled with a forward pitch motion and acceleration mechanisms provided herein can determine that a kick event is occurring.

[0090] In some examples, the change in vehicle component positioning, change in vertical acceleration, and/or change in pitch rate are compared to one or more thresholds to determine whether to implement an adjustment of a damping characteristic to one or more adjustable shock absorbers of the vehicle. Therefore, by comparing the changes to such thresholds, mechanisms provided herein may ensure that adjustments to damping characteristics of adjustable shock absorbers are not made during relatively normal driving conditions, but rather only in rear unloading events.

[0091] In some examples, the vertical acceleration is multiplied by the pitch rate to determine a rear unloading event score and/or magnitude. In some examples, a pitch acceleration (e.g., rate of change of a pitch rate) is determined and used in calculations discussed herein, such as instead of the pitch rate. In some examples, the rear unloading event score is compared to a threshold to determine whether to implement an adjustment of a damping characteristic to one or more adjustable shock absorbers of the vehicle. Additional and/or alternative calculations, which use and/or are based on the changes in acceleration, vehicle component positioning, and/or pitch rate, to calculate the rear unloading event score, should be recognized by those of ordinary skill in the art.

[0092] In some examples, the rear unloading event score, change in acceleration, vehicle component positioning, and/or change in pitch rate may be compared to a look-up table to determine whether to adjust a damping characteristic of one or more adjustable shock absorbers of the vehicle. In some examples, the comparison to the look-up table may be used to determine what type of damping characteristic (e.g., compression damping and/or rebound damping) to adjust. In some examples, the comparison to the look-up table may be used to determine how much to adjust the damping characteristic of the one or more adjustable shock absorbers of the vehicle.

[0093] In some examples, the thresholds to which the changes in vertical acceleration, vehicle component positioning, and/or pitch rate are compared may be calibrated. For example, the thresholds may be calibrated based on and/or correspond to a type of vehicle (e.g., vehicle model), mode of vehicle (e.g., ECO mode, sport mode, work mode, two-wheel drive, four-wheel drive, etc.), operator preference, manufacture preference, historical data of vehicle component positioning, acceleration and/or pitch information (e.g., vertical acceleration and pitch rate) collected over a duration of time (e.g., hours, days, months years), or other factors which should be recognized by those of ordinary skill in the art. Similarly, in some examples, the look-up table may be calibrated. For example, the look-up table may be calibrated based on and/or correspond to a type of vehicle (e.g., vehicle model), mode of vehicle (e.g., ECO mode, sport mode, work mode, two-wheel drive, four-wheel drive, etc.), operator preference, manufacture preference, historical data of vehicle component positioning, acceleration and/or pitch information (e.g., vertical acceleration and pitch rate) collected over a duration of time (e.g., hours, days, months years), or other factors which should be recognized by those of ordinary skill in the art.

[0094] In some examples, the control system 300 functions continuously, such that adjustments to damping characteristics of one or more shock absorbers of the vehicle are made smoothly, rather than as a binary on/off function. In some examples, the vertical acceleration and/or change in pitch rate are received from an IMU, such as the IMU 37 of FIG. 10. In some examples, the vehicle component positioning measurements are received from a shock absorber position sensor (e.g., such as when measuring the position of shock absorbers). In some examples, the vehicle component positioning measurements are received from another vehicle component positioning

sensor corresponding to one or more particular vehicle components. In some examples, the control system **300** may be in communication with other types of sensors, such as wheel acceleration sensors, shock acceleration sensors, axle sensors, camera sensors, radar sensors, LIDAR sensors, or any other sensors recognized by those of ordinary skill in the art which may be used to detect when vehicle suspension is going to compress or unload. Accordingly, in some examples, the control system **300** can transmit one or more signals to enable adjustments to one or more adjustable shock absorbers, in addition to or independent of kinematic sensors of a vehicle.

[0095] FIG. **14** shows an example flowchart describing the operation of the suspension controller **86** during a rear unloading event. The flowchart illustrates an example method or process **500**. In some examples, the method **500** is a method of adjusting one or more adjustable shock absorbers. As will be described in more detail below, by adjusting the rebound damping of one or more adjustable shock absorbers **18** due to a rear unloading event, the suspension controller **86** may prevent the rear end of the vehicle **10** from launching upward and/or forward with respect to the front end of the vehicle **10**.

[0096] In operation, at step **502**, the suspension controller **86** may receive information (e.g., inputs) from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive (e.g., retrieve and/or obtain) information (e.g., data packets and/or signals indicating sensor readings) from the one or more sensors, devices, and/or subsystems. In some instances, the suspension controller **86** may receive information indicating the x, y, and/or z-axis acceleration from the chassis accelerometer **30** and/or IMU **37**. For example, referring back to FIG. **3**, the chassis accelerometer **30** and/or IMU **37** may measure the x, y, and/or z-axis acceleration values for the vehicle **10**, and may provide the acceleration values to the suspension controller **86**. Further, the suspension controller **86** may receive information indicating a pitch rate of the vehicle, such as from the IMU **37**. Still further, the suspension controller **86** may receive information indicating a position of a component of the vehicle, such as from a shock absorber positioning sensor and/or a positioning sensor configured to monitor the position of another vehicle component.

[0097] At step **506**, the suspension controller **86** may determine whether a rear unloading event is occurring. For example, the suspension controller **86** may determine whether a rear unloading event occurring based on the z-axis acceleration value (i.e., vertical acceleration), pitch rate of the vehicle, and or positioning of one or more vehicle components. For instance, the suspension controller **86** may compare the z-axis acceleration and pitch rate values with one or more thresholds (e.g., one or more pre-defined, pre-programmed, and/or user-defined thresholds) to determine whether the vehicle **10** is in a rear unloading event. If the z-axis acceleration value and pitch rate value are greater than the one or more thresholds, the method **500** may move to step **508**. If the z-axis acceleration value and pitch rate value are less than the thresholds, the method **500** may move back to step **502** and repeat.

[0098] In some examples, the vertical acceleration is run through a low-pass filter (e.g., at 0.25 hertz). In some examples, running the vertical acceleration through a low-pass filter gives a delayed vertical acceleration signal. In some examples, the low-pass filtering is helpful, such as in cases where the vertical acceleration is dropping due to the vehicle becoming airborne. In some examples, the low-pass filter also prevents mechanisms from inadvertently triggering. In some examples, prior to performing calculations using the vertical acceleration and/or pitch acceleration, mechanisms provided herein verify that the vertical acceleration was/is greater than 1 and that the pitch acceleration is in a positive rotational direction (e.g., as illustrated by the rotational arrows pointing clockwise in FIG. **13**).

[0099] In some examples, the suspension controller **86** may use any of a plurality of different techniques for comparing the z-axis acceleration and pitch rate to thresholds. In some examples, the z-axis acceleration is multiplied by the pitch rate to determine a rear unloading event score and/or magnitude. In some examples, the rear loading event score is compared to a threshold to determine

whether to implement an adjustment of a damping characteristic to one or more adjustable shock absorbers of the vehicle. Additional and/or alternative calculations, which use and/or are based on the changes in acceleration and pitch rate, to calculate the rear unloading event score, should be recognized by those of ordinary skill in the art.

[0100] In some examples, the rear unloading event score, change in acceleration, and/or change in pitch rate may be compared to a look-up table to determine whether to adjust a damping characteristic of one or more adjustable shock absorbers of the vehicle. In some examples, the comparison to the look-up table may be used to determine what type of damping characteristic (e.g., compression damping and/or rebound damping) to adjust. In some examples, the comparison to the look-up table may be used to determine how much to adjust the damping characteristic of the one or more adjustable shock absorbers of the vehicle.

[0101] At step **508**, the suspension controller **86** may determine a rear unloading event modifier for one or more of the adjustable shock absorbers **18**. For example, the suspension controller **86** may provide information (e.g., one or more commands) to the adjustable shock absorbers **18** to increase and/or gradually increase the rebound damping. By increasing the adjustable shock absorbers **18**, the vehicle **10** may prevent or reduce the extent to which the rear of the vehicle unloads and/or pitches forward with respect to the front of the vehicle.

[0102] At step **510**, the suspension controller **86** may determine whether a rear unloading event has occurred and/or is done/completed. For example, similar to step **506**, the suspension controller **86** may determine whether the rear unloading event is done based on the z-axis acceleration value and/or pitch rate of the vehicle. For instance, the suspension controller **86** may compare the z-axis acceleration value and/or pitch rate with one or more thresholds (e.g., one or more pre-defined, pre-programmed, and/or user-defined thresholds) to determine whether the vehicle **10** is no longer in a rear unloading event. In some examples, the suspension controller **86** may use the same thresholds as in step **506**.

[0103] At step **512**, the suspension controller **86** may execute a normal condition modifier. For example, the suspension controller **86** may adjust and/or gradually adjust the rebound damping back to normal (e.g., back to the pre-rear unloading event position).

[0104] In some examples, the method **500** terminates at step **512**. Alternatively, in some examples, the method **500** returns to step **502**, from step **512**, such as to provide a continuous loop of vehicle information (e.g., acceleration information and/or pitch information) to determine whether, and by how much, to adjust a damping characteristic of one or more adjustable shock absorbers. Thus, while method **400** depicts operation **408** as executing a rear loading event modifier and operation **412** as executing a normal condition modifier, it will be appreciated that, in some instances, similar aspects may be used to gradually adjust rebound damping of a vehicle between such modifiers (e.g., gradually and/or as a combination thereof).

[0105] While the rear unloading event and rear loading event have been described in detail above, one of ordinary skill in the art should recognize that the suspension control system **300** may be used to operate the suspension controller **86** during other types of events and/or operations. For example, other types of events and/or operations with which the suspension control system **300** may be used include a breaking event, cornering event, airborne event, hill climbing event, hill sliding event, rock crawler operation, and/or other types of events/operations that may be recognized by those of ordinary skill in the art, such as discussed in U.S. Pat. No. 11,904,648, entitled ADJUSTABLE SUSPENSIONS AND VEHICLE OPERATION FOR OFF-ROAD RECREATIONAL VEHICLES (filed on Jul. 19, 2021), and U.S. Pat. No. 10,987,987, entitled VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING (filed Nov. 21, 2018), which are assigned to the present assignee and the entire disclosures are expressly incorporated by reference herein.

[0106] FIG. **15** shows an example of the suspension controller **86** adjusting the shock absorbers **18** for the vehicle **10** during a rear loading or rear unloading event. In some examples, for better

performance during a rear loading and/or unloading event, a vehicle, such as vehicle **10**, may use a controller **20** (e.g., the suspension controller **86**) to adjust the rebound and/or compression damping. For instance, during a rear loading event, the suspension controller **86** may adjust (e.g., increase) the compression damping of the shock absorbers **18** (e.g., rear shock absorbers **18c**, **18d**), which may cause the vehicle **10** to not bottom out (e.g., at all or to a reduced degree) during a rear loading event. In some examples, during a rear unloading event, the suspension controller **86** may adjust (e.g., increase) the rebound damping of the shock absorber **18** (e.g., rear shock absorbers **18c**, **18d**), which may cause the rear end of the vehicle to not get launched up and/or forward into the air (e.g., at all or to a reduced degree), relative to the front end of the vehicle.

[0107] FIG. **16** shows a schematic block diagram illustrating an example of logic components in the suspension controller **86**. For example, the suspension controller **86** includes one or more logic components, such as an acceleration and/or velocity measurement logic **720**, angular acceleration logic **722**, angular acceleration calculation logic **724**, kinematics function logic **726**, estimated acceleration at center of gravity (CG) logic **728**, offset correction based on user or cargo information logic **730**, and/or offset correction based on sensor information logic **732**. The logic **720**, **722**, **724**, **726**, **728**, **730**, and/or **732** is any suitable logic configuration including, but not limited to, one or more state machines, processors that execute kernels, and/or other suitable structure as desired.

[0108] Further, in some examples, the suspension controller **86** and/or the controller **20** may include memory and one or more processors. The memory may store computer-executable instructions that when executed by the one or more processors cause the processors to implement one or more methods and/or procedures discussed herein. Additionally, various components (e.g., logic) depicted in FIG. **16** are, in embodiments, integrated with various ones of the other components depicted therein (and/or components not illustrated), all of which are considered to be within the ambit of the present disclosure. In some embodiments, various components depicted in FIG. **16** may be integrated with components discussed earlier herein with respect to other figures of the present disclosure.

[0109] In operation, at step **702**, the suspension controller **86** (e.g., logic **720**) may receive vehicle movement information from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive vehicle movement information indicating the acceleration and/or velocity of the vehicle **10** from one or more sensors. For instance, the logic **720** may receive x, y, and/or z acceleration values (e.g., linear acceleration values) from a sensor, such as the IMU **37** and/or the chassis accelerometer **30**. The logic **720** may also receive x, y, and/or z angular velocity values from a sensor, such as the IMU **37** and/or the gyroscope **31**. The logic **720** may also receive pitch values from the IMU **37** and/or another sensor.

[0110] At step **704**, the suspension controller **86** (e.g., logic **730**) may receive user or cargo information from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. In some instances, the suspension controller **86** may receive user or cargo information from the operator interface **22**. For example, the operator interface **22** may provide, to the suspension controller **86**, user information indicating a number of riders in the vehicle **10** and/or a weight or mass corresponding to the riders in the vehicle **10**. Additionally, and/or alternatively, the operator interface **22** may provide, to the suspension controller **86**, cargo information indicating a weight or mass of cargo that the vehicle **10** is carrying.

[0111] In some examples, the suspension controller **86** may receive the user or cargo information from one or more sensors. For example, one or more sensors, such as seat belt sensors, may provide, to the suspension controller **86**, information indicating the number of riders in the vehicle **10** (e.g., when a user puts on their seat belt, the seat belt sensor may provide information indicating the seat belt is engaged to the suspension controller **86**).

[0112] At step **706**, the suspension controller **86** (e.g., the logic **722** and/or **724**) may determine the angular acceleration for the vehicle **10**. For example, the logic **722** may filter the angular velocity

and/or rates received from a sensor, such as the IMU **37** and/or gyroscope **31**. The logic **724** may differentiate the angular rates to determine the angular acceleration.

[0113] At step **708**, the suspension controller **86** (e.g., the logic **732**) may determine a first offset correction based on a distance between a sensor (e.g., the IMU **37**, the chassis accelerometer **30**, and/or the gyroscope **31**) and the CG of the vehicle **10**. The CG of the vehicle **10** may be pre-programmed and/or pre-defined. Additionally, and/or alternatively, the suspension controller **86** (e.g., the logic **732**) may determine the first offset correction based on a distance between a sensor (e.g., the IMU **37**, the chassis accelerometer **30**, and/or the gyroscope **31**) and the suspension controller **86**.

[0114] At step **710**, the suspension controller **86** (e.g., the logic **730**) may determine a second offset correction based on the user and/or cargo information received at step **704**. For example, based on the number of users, weight/mass of users, and/or weight/mass of the cargo, the logic **730** may determine a second offset correction.

[0115] At step **712**, the suspension controller **86** (e.g., the logic **726** and/or **728**) may determine an estimated acceleration at the CG of the vehicle **10**. For example, the logic **726** may receive information from the logic **720** (e.g., the linear acceleration values), the logic **724** (e.g., the determined angular acceleration values), the logic **730** (e.g., the second offset based on the user or cargo information), and/or the logic **732** (e.g., the first offset based on the sensor information). The logic **726** may use a kinematics function to determine the 3-axis estimated acceleration at the CG of the vehicle **10**. The logic **726** may provide the 3-axis estimated acceleration at the CG of the vehicle **10** to the logic **728**. Then, the method **700** may return to step **702**, and may repeat continuously.

[0116] In some examples, the logic **726** may receive and/or store the 3-axis (e.g., x, y, and/or z-axis) estimated acceleration values at the CG of the vehicle **10**. Additionally, and/or alternatively, the methods disclosed throughout this disclosure may use the determined estimated acceleration values at the CG of the vehicle **10** to execute rear unloading event modifiers, rear loading event modifiers, and/or other event or operation modifiers which may be recognized by those of ordinary skill in the art. For example, the suspension controller **86** may use the estimated acceleration values at the CG of the vehicle **10** to determine and/or detect events (e.g., rear loading events, rear unloading events, etc.). Additionally, and/or alternatively, the suspension controller **86** may use the estimated acceleration values at the CG of the vehicle **10** to adjust the compression damping and/or rebound damping of the adjustable shock absorbers **18**.

Stabilizer Bar

[0117] FIG. **17** illustrates a representative view of a stabilizer bar assembly **1004** of a vehicle, such as vehicle **10**. In embodiments, vehicle **10** may include stabilizer bar **1004** coupled between the rear suspension (e.g., rear suspension **216A**, **216B**) and frame **14**. In embodiments, stabilizer bar assembly **1004** includes a stabilizer bar **1005** and one or more links **1010**. In embodiments, stabilizer bar assembly **1004** is an adjustable stabilizer bar. In embodiments, the one or more links **1010** are adjustable links with adjustable suspension characteristics (e.g., locking status, damping characteristic). In embodiments, the one or more links **1010** are adjustable shock absorbers with adjustable suspension characteristics (e.g., locking status, damping characteristic). In embodiments, the links **1010** are operably coupled to a controller (e.g., suspension controller **86**).

[0118] In embodiments, a first link **1010** may be coupled between a first end **1007** of the stabilizer bar **1005** and a first trailing arm or control arm **1016** of rear suspension **216A** and a second link (not shown, similar to or the same as first link **1010**) may be coupled between a second end (not shown, similar to first end **1007**) of stabilizer bar **1005** and a second trailing arm or control arm (not shown, similar to control arm **1016**) of rear suspension **216B**. In embodiments, a control arm or steering arm **1015** extends between frame **14** and a ground engaging member.

[0119] In embodiments, the one or more links **1010** are adjustable links including a first adjustable link **1010** coupled between the first end **1007** of the stabilizer bar **1005** and the first trailing arm or

control arm **1016** of rear suspension **216A** and the second adjustable link (not shown, similar to or the same as first link **1010**) may be coupled between the second end (not shown, similar to first end **1007**) of stabilizer bar **1005** and the second trailing arm or control arm (not shown, similar to control arm **1016**) of rear suspension **216B**

[0120] The stabilizer bar assembly **1004** may be in the rear of the vehicle, and located close to a rear adjustable shock absorber, such as shock absorber **18c**. In embodiments, stabilizer bar assembly **1004** may be positioned in a front of the vehicle and coupled between front suspensions **214A**, **214B**.

[0121] In embodiments, adjustable links **1010** may be capable of locking (e.g., maintaining a static position) or unlocking (e.g., allowing movement to increase or decrease the length of adjustable link **1010**). In embodiments, adjustable links **1010** may be capable of operating at discrete damping levels (e.g., compression and/or rebound damping). In embodiments, adjustable links **1010** may be capable of operating at an infinite (e.g., continuous) number of damping levels (e.g., compression and/or rebound damping). In embodiments, a controller (e.g., suspension controller **86**) may send and/or receive signals related to damping characteristics, locking characteristics, or other characteristics to adjustable links **1010**. For example, the controller can actively control the adjustable links **1010**, such as in addition to controlling the shock absorbers **18**.

[0122] Also, similar to the adjustable shock absorbers **18**, the suspension controller **86** may provide one or more commands to adjust the damping characteristics of the stabilizer bar adjustable shock absorbers **1010**. In embodiments, suspension controller **86** may provide one or more commands to adjust a locking status (e.g., locked or unlocked) of the one or more adjustable links **1010**. In embodiments, in response to detecting a rear loading event and/or a rear unloading event, suspension controller **86** may provide one or more commands to adjust a damping characteristic to the stabilizer bar assembly **1004**. In embodiments, in response to a rear loading event and/or a rear unloading event, suspension controller **86** may provide one or more commands to adjust a locking status of the stabilizer bar assembly **1004**. In some embodiments, the stabilizer bar assembly **1004** has two states (e.g., open or locked). For example, the adjustable links **1010** can be open or locked. In some embodiments, an increase in damping for the stabilizer bar assembly **1004** while in an open state has variable damping (e.g., variable damping of the adjustable links **1010**).

[0123] In embodiments, suspension controller **86** may simultaneously provide one or more commands to adjustable links **1010** and one or more commands to adjustable shock absorbers **18a**, **18b**, **18c**, **18d**. In embodiments, in response to a rear loading event and/or a rear unloading event, suspension controller **86** may provide one or more commands to adjust a suspension characteristic of the stabilizer bar assembly **1004** and a suspension characteristic of one or more of adjustable shock absorbers **18a**, **18b**, **18c**, **18d**. In embodiments, in response to a rear loading event and/or a rear unloading event, suspension controller **86** may provide a command to increase or decrease the stiffness of stabilizer bar assembly **1004**. In embodiments, the stiffness of stabilizer bar assembly **1004** may be increased by locking one or more of links **1010** or by increasing a compression damping of one or more of links **1010**. In embodiments, the stiffness of stabilizer bar assembly **1004** may be decreased by unlocking one or more of links **1010** or by decreasing a compression damping of one or more of links **1010**.

[0124] Exemplary stabilizer bars are described in U.S. Pat. No. 9,365,251 (filed Jun. 14, 2016, entitled SIDE-BY-SIDE VEHICLE), and U.S. Provisional Patent Application No. 63/528,547 (filed Jul. 24, 2023, entitled ADJUSTABLE SUSPENSION ASSEMBLIES AND METHODS OF USE THEREOF) which are assigned to the present assignee and the entire disclosures of which are expressly incorporated by reference herein.

[0125] FIG. **18** illustrates a view of an exemplary vehicle **1100**, such as a snowmobile. Vehicle **1100**, as illustrated, includes a plurality of ground engaging members **12**. Illustratively, the ground engaging members **12** are the endless track assembly **1108** and a pair of front skis **1112a** and **1112b**. The endless track assembly **1108** is operatively coupled to power system **60** (see FIG. **4**) to power

the movement of vehicle **1100**. The vehicle may also include a seat **1102** and a seat surface **1104**. Also, the vehicle may include handlebars **1106**.

[0126] Further, the suspensions **1114** and **1116** are coupled to the frame of the vehicle **1100** and the pair of front skis **1112a** and **1112b**. The suspensions **1114** and **1116** may include adjustable shock absorbers, such as the adjustable shock absorber **1110**. Also, the endless track assembly **1108** may also be coupled to one or more suspensions and/or adjustable shock absorbers.

[0127] The vehicle **1100** may be the same and/or include components of vehicle **10**. For example, the vehicle **1100** may include the plurality of vehicle condition sensors **40** described above, and may also include one or more controllers, such as the suspension controller **86**. The suspension controller **86** may receive information from the plurality of vehicle condition sensors **40**. Using the received information, the suspension controller **86** may adjust the compression and/or rebound damping of the adjustable shock absorbers, such as adjustable shock absorber **1110** as described above. Additional details regarding vehicle **1100** are provided in U.S. Pat. Nos. 9,809,195 and 8,994,494, assigned to the present assignee, the entire disclosures of which are expressly incorporated by reference herein. Further, additional details regarding damping control for shock absorbers of the vehicle **1100** are provided in US Patent Pub. No. 2021/0362806, entitled SYSTEMS AND METHODS OF ADJUSTABLE SUSPENSIONS FOR OFF-ROAD RECREATIONAL VEHICLES (filed May 19, 2021), assigned to the present assignee, the entire disclosures of which is expressly incorporated by reference herein

[0128] In embodiments, substantially zero is any value which is effectively zero. For example, a substantially zero value does not provide an appreciable difference in the operation compared to when the value is zero. Aspects of the present disclosure, for example, are described above with reference to block diagrams and/or operational illustrations of methods, systems, and computer program products according to aspects of the disclosure. The functions/acts noted in the blocks may occur out of the order as shown in any flowchart. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

[0129] The following clauses illustrate example subject matter described herein.

[0130] Clause 1: A recreational vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers; at least one sensor positioned on the recreational vehicle and configured to provide one or more of acceleration information, pitch information, or position information to a controller; and the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to: receive, from the at least one sensor, the one or more of acceleration information, pitch information, or position information; determine, based on the one or more of acceleration information, pitch information, or position information, to adjust damping of at least one of the plurality of adjustable shock absorbers; and provide, to the at least one of the plurality of adjustable shock absorbers, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

[0131] Clause 2: The recreational vehicle of claim **1**, further comprising a stabilizer bar coupled between a rear suspension of the plurality of suspensions and the frame.

[0132] Clause 3: The recreational vehicle of any of the preceding clauses, wherein the recreational vehicle is a snowmobile, and wherein the plurality of ground engaging members include skis.

[0133] Clause 4: The recreational vehicle of any of the preceding clauses, wherein the controller is further configured to: determine, based on the one or more of acceleration information, pitch information, and position information, that a rear loading event is occurring, wherein the one or more commands comprise a command to increase a compression damping characteristic of the at

least one of the plurality of adjustable shock absorbers.

[0134] Clause 5: The recreational vehicle of clause 4, wherein, to determine that the rear loading event is occurring, a rear loading event score is calculated based on the one or more of acceleration information, pitch information, and position information, and the increase in the compression damping characteristic is based on the magnitude of the rear loading event score.

[0135] Clause 6: The recreational vehicle of any one of clauses 1-3, wherein the controller is further configured to: determine, based on the one or more of acceleration information, pitch information, and position information that a rear unloading event is occurring, wherein the one or more commands comprise a command to increase a rebound damping characteristic of the at least one of the plurality of adjustable shock absorbers.

[0136] Clause 7: The recreational vehicle of clause 6, wherein, to determine that the rear unloading event is occurring, a rear unloading event score is calculated based on the one or more of acceleration information, pitch information, and positions information, and the increase in the rebound damping characteristic is based on the magnitude of the rear unloading event score.

[0137] Clause 8: A method of adjusting one or more adjustable shock absorbers of a vehicle, the method comprising: receiving, from at least one sensor of the vehicle, acceleration information and pitch information; determining, based on the acceleration information and pitch information, to adjust damping of at least one of the one or more adjustable shock absorbers; and providing, to the at least one of the one or more adjustable shock absorbers, one or more commands to cause an adjustment of a damping characteristic of the at least one of the one or more adjustable shock absorbers.

[0138] Clause 9: The method of clause 8, wherein the at least one sensor comprises at least one of: an accelerometer or an inertial measurement unit (IMU).

[0139] Clause 10: The method of clause 8 or 9, wherein the damping characteristic comprises at least one of: a compression damping or a rebound damping.

[0140] Clause 11: The method of any one of clauses 8-10, further comprising: determining, based on the acceleration information and pitch information, that a rear loading event is occurring, by: calculating a rear loading event score based on the acceleration information and pitch information; and comparing the rear loading event score to one or more thresholds, wherein the one or more commands comprise a command to increase a compression damping characteristic of the at least one of the one or more adjustable shock absorbers.

[0141] Clause 12: The method of clause 11, wherein the one or more thresholds correspond to one or more of: a vehicle type, a vehicle mode, or historical acceleration and pitch information data.

[0142] Clause 13: The method of any one of clauses 8-10, further comprising: determining, based on the acceleration information and pitch information, that a rear unloading event is occurring, wherein the one or more commands comprise a command to increase a rebound damping characteristic of the at least one of the one or more adjustable shock absorbers.

[0143] Clause 14: The method of clause 13, wherein, to determine that the rear unloading event is occurring, a rear unloading event score is calculated based on the acceleration information and pitch information, and the increase in the rebound damping characteristic is based on the rear unloading event score.

[0144] Clause 15: A recreational vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame and includes an adjustable shock absorber of a plurality of adjustable shock absorbers; at least one sensor positioned on the recreational vehicle, the at least one sensor configured to provide acceleration information and pitch information to a controller; and the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to: receive, from the at least one sensor, the acceleration information and the pitch information; calculate an event score based on the

acceleration information and the pitch information; compare the calculated event score to a threshold; and control at least one adjustable shock absorber of the plurality of adjustable shock absorbers to adjust a damping characteristic of the at least one adjustable shock absorber based on the comparison.

[0145] Clause 16: The recreational vehicle of clause 15, wherein the at least one sensor comprises at least one of: an accelerometer or an inertial measurement unit (IMU).

[0146] Clause 17: The recreational vehicle of clause 15 or 16, wherein the damping characteristic comprises at least one of: a compression damping or a rebound damping.

[0147] Clause 18: The recreational vehicle of any one of clauses 15-17, wherein controlling the at least one adjustable shock absorber comprises generating a command to increase the damping characteristic of the at least one of the plurality of adjustable shock absorbers.

[0148] Clause 19: The recreational vehicle of any one of clauses 15-18, wherein the acceleration information comprises a vertical acceleration of the recreational vehicle and wherein the score is calculated based on the vertical acceleration and the pitch information.

[0149] Clause 20: The recreational vehicle of any one of clauses 15-19, wherein the threshold corresponds to one or more of: a vehicle type, a vehicle mode, or historical acceleration and pitch information data.

[0150] The above detailed description of the present disclosure and the examples described therein have been presented for the purposes of illustration and description only and not by limitation. It is therefore contemplated that the present disclosure covers any and all modifications, variations or equivalents that fall within the scope of the basic underlying principles disclosed above and claimed herein.

Claims

1. A recreational vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers; at least one sensor positioned on the recreational vehicle and configured to provide one or more of acceleration information, pitch information, or position information to a controller; and the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to: receive, from the at least one sensor, the one or more of acceleration information, pitch information, or position information; determine, based on the one or more of acceleration information, pitch information, or position information, to adjust damping of at least one of the plurality of adjustable shock absorbers; and provide, to the at least one of the plurality of adjustable shock absorbers, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

2. The recreational vehicle of claim 1, further comprising a stabilizer bar coupled between a rear suspension of the plurality of suspensions and the frame.

3. The recreational vehicle of claim 1, wherein the recreational vehicle is a snowmobile, and wherein the plurality of ground engaging members include skis.

4. The recreational vehicle of claim 1, wherein the controller is further configured to: determine, based on the one or more of acceleration information, pitch information, and position information, that a rear loading event is occurring, wherein the one or more commands comprise a command to increase a compression damping characteristic of the at least one of the plurality of adjustable shock absorbers.

5. The recreational vehicle of claim 4, wherein, to determine that the rear loading event is occurring, a rear loading event score is calculated based on the one or more of acceleration

information, pitch information, and position information, and the increase in the compression damping characteristic is based on the magnitude of the rear loading event score.

6. The recreational vehicle of claim 1, wherein the controller is further configured to: determine, based on the one or more of acceleration information, pitch information, and position information that a rear unloading event is occurring, wherein the one or more commands comprise a command to increase a rebound damping characteristic of the at least one of the plurality of adjustable shock absorbers.

7. The recreational vehicle of claim 6, wherein, to determine that the rear unloading event is occurring, a rear unloading event score is calculated based on the one or more of acceleration information, pitch information, and positions information, and the increase in the rebound damping characteristic is based on the magnitude of the rear unloading event score.

8. A method of adjusting one or more adjustable shock absorbers of a vehicle, the method comprising: receiving, from at least one sensor of the vehicle, acceleration information and pitch information; determining, based on the acceleration information and pitch information, to adjust damping of at least one of the one or more adjustable shock absorbers; and providing, to the at least one of the one or more adjustable shock absorbers, one or more commands to cause an adjustment of a damping characteristic of the at least one of the one or more adjustable shock absorbers.

9. The method of claim 8, wherein the at least one sensor comprises at least one of: an accelerometer or an inertial measurement unit (IMU).

10. The method of claim 8, wherein the damping characteristic comprises at least one of: a compression damping or a rebound damping.

11. The method of claim 8, further comprising: determining, based on the acceleration information and pitch information, that a rear loading event is occurring, by: calculating a rear loading event score based on the acceleration information and pitch information; and comparing the rear loading event score to one or more thresholds, wherein the one or more commands comprise a command to increase a compression damping characteristic of the at least one of the one or more adjustable shock absorbers.

12. The method of claim 11, wherein the one or more thresholds correspond to one or more of: a vehicle type, a vehicle mode, or historical acceleration and pitch information data.

13. The method of claim 8, further comprising: determining, based on the acceleration information and pitch information, that a rear unloading event is occurring, wherein the one or more commands comprise a command to increase a rebound damping characteristic of the at least one of the one or more adjustable shock absorbers.

14. The method of claim 13, wherein, to determine that the rear unloading event is occurring, a rear unloading event score is calculated based on the acceleration information and pitch information, and the increase in the rebound damping characteristic is based on the rear unloading event score.

15. A recreational vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame and includes an adjustable shock absorber of a plurality of adjustable shock absorbers; at least one sensor positioned on the recreational vehicle, the at least one sensor configured to provide acceleration information and pitch information to a controller; and the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to: receive, from the at least one sensor, the acceleration information and the pitch information; calculate an event score based on the acceleration information and the pitch information; compare the calculated event score to a threshold; and control at least one adjustable shock absorber of the plurality of adjustable shock absorbers to adjust a damping characteristic of the at least one adjustable shock absorber based on the comparison.

16. The recreational vehicle of claim 15, wherein the at least one sensor comprises at least one of:

an accelerometer or an inertial measurement unit (IMU).

17. The recreational vehicle of claim 15, wherein the damping characteristic comprises at least one of: a compression damping or a rebound damping.

18. The recreational vehicle of claim 17, wherein controlling the at least one adjustable shock absorber comprises generating a command to increase the damping characteristic of the at least one of the plurality of adjustable shock absorbers.

19. The recreational vehicle of claim 15, wherein the acceleration information comprises a vertical acceleration of the recreational vehicle and wherein the score is calculated based on the vertical acceleration and the pitch information.

20. The recreational vehicle of claim 15, wherein the threshold corresponds to one or more of: a vehicle type, a vehicle mode, or historical acceleration and pitch information data.
